Unified Master Compendium: Dimensionless Invariants and Semiclassical Diagnostic Framework (v2.0)

Luís Pimenta Damásio
Independent Researcher, Lisbon, Portugal • ORCID: 0009-0001-7606-7557
Master Compendium DOI: 10.5281/zenodo.24163322 • Series Scope: Papers 1–13

November 2026

AI COLLABORATION & AUTHORSHIP TRANSPARENCY STATEMENT
This theoretical framework, its mathematical derivations, and its manuscript preparation were developed through an interactive human–AI research collaboration between **Luís Pimenta Damásio** and Google's **Gemini** AI assistant. Conceptual direction, physical synthesis, metric architecture, and final editorial verification were directed by the human author, with formal mathematical derivations, LaTeX typesetting, and literature verification generated with the aid of Gemini.

Abstract

This Master Compendium (v2.0) unifies a four-part mathematical foundation and ten thematic research monographs (Papers 1–13) establishing a covariant, dimensionless invariant architecture across General Relativity, Quantum Field Theory in Curved Spacetime (QFTCS), Open Quantum Systems, and Relativistic Quantum Metrology. By formulating physical observables in terms of Zero-Angular-Momentum Observer (ZAMO) proper-time exchange rates (ρ_τ), Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) decoherence actions (Ω) bounded by Bekenstein–Hawking Hilbert-space floors (Φ_{floor}), Seeley–DeWitt heat-kernel conformal purities (χ), an L_∞ Chebyshev semiclassical breakdown regulator (Ξ_{QG}), and Coherence-Bounded Operational Locality metrics ($\mathcal{L}_{\text{sim}}$, $\mathcal{S}_{\text{sim}}$, $\mathcal{D}_{\text{loc}}$, $\Lambda_{A\to B}$, V_{null}, $\mathcal{L}_{\text{sim}}(r)$), this framework bridges microphysical quantum phase coherence, macroscopic horizon thermodynamics, and finite-latency relativistic synchronization across 120 orders of magnitude within a neutral four-dimensionalist ontology.

Keywords: Semiclassical Gravity, Quantum Field Theory in Curved Spacetime, ZAMO Foliation, Open Quantum Systems, Bekenstein–Hawking Holography, Quantum Extremal Surfaces, Superradiance, Operational Simultaneity Horizons, Coherence-Bounded Locality, Quantum Reference Frames.

1 Master Table of Contents (Papers 1–13)

Part I: Foundational Custom-Metric Specifications

1. **Covariant Chronological Kinematics and ZAMO Frame Foliation:** ADM 3+1 and Boyer–Lindquist axisymmetric foliation, ZAMO normal congruence n^μ, relational proper-time dilation ρ_τ, clock rate delta $\Delta\rho_\tau$, wave-crest redshift z, and quantum frequency suppression H_Q.

2. **Open-System Decoherence and Holographic Phase Dynamics:** GKSL master equation dynamics, Caldeira–Leggett thermal scattering Γ_{th}, Penrose–Diósi gravitational self-energy dephasing Γ_{grav}, decoherence action Ω, Bekenstein–Hawking inverse-dimension floor Φ_{floor}, Timeline Integrity Φ_τ, surviving coherent cycles N_{eff}, and the Lloyd–Bekenstein gate bound η_{Lloyd}.

3. **Geometric Purity and Multi-Scale Diagnostic Ratios:** Seeley–DeWitt a_4 trace anomaly normalization and Weyl Conformal Purity χ, Buckingham-Π gravity-to-thermal crossover R_{DP}, Penrose–Zel'dovich rotational extraction gain $E_{\text{Penrose}}(\theta)$, and the L_∞ Semiclassical Breakdown Index Ξ_{QG}.

4. **Operational Simultaneity Horizons and Coherence-Bounded Locality (v2.0 Expansion):** Two-way Einstein–Poincaré round-trip signal delay $\Delta\tau_{\text{sig}}$ across refractive waveguides (n_{eff}), Operational Simultaneity Horizon $\mathcal{L}_{\text{sim}}$, Operational Simultaneity Fidelity $\mathcal{S}_{\text{sim}}(\Omega, \Delta\ell)$, post-selected CHSH Bell correlations $S_{\text{CHSH}}(\Omega)$, and the Spatial Locality Decay Factor $\mathcal{D}_{\text{loc}}$.

Part II: Thematic Research Monographs (Papers 4–13)

1. **Unitary Black Hole Evaporation, Quantum Extremal Surfaces, and the Page Curve:** Ingoing Vaidya collapse, Bogoliubov mode peeling, Quantum Extremal Surface (Island) extremization, Euclidean replica wormhole saddle exchange, and Kerr–Newman evaporation bounded by Φ_{floor} and Ξ_{QG}.

2. **The Quantum-to-Classical Transition: Environmental Decoherence and Gravitationally Induced Collapse:** Feynman–Vernon influence functionals, Itô stochastic localization, and the $R_{\text{DP}} = 1$ mesoscopic crossover between thermal scattering and gravitational self-energy collapse.

3. **Relational Time and Covariant Foliations in Relativistic Quantum Mechanics:** Dirac–Bergmann constraints, the Wheeler–DeWitt frozen formalism, Page–Wootters conditional probability states, Tomonaga–Schwinger multi-fingered time evolution, and ZAMO worldline synchronization.

4. **Trans-Planckian and Curvature Singularity Regulation in Semiclassical Gravity:** Hadamard point-splitting renormalization, Schwinger proper-time heat kernels, Kempf–Mangano–Mann GUP minimal-length cutoffs, Corley–Jacobson dispersive mode bounce, and Ξ_{QG} horizon membrane regulation.

5. **Rotational Energy Extraction and Ultralight Dark Matter Superradiance:** Newman–Penrose tetrad decoupling, the Teukolsky master equation, Sturm–Liouville Wronskian conservation, Zel'dovich–Starobinsky amplification, and Detweiler's gravitational-atom instability scaling $(\alpha_G, \tau_{\text{inst}})$.

6. **Operational Simultaneity Horizons and the Resolution of the Andromeda Paradox (Paper 9):** Derivation of $\mathcal{L}_{\text{sim}}$ and proof that macroscopically separated events lie beyond active phase-locked simultaneity, resolving Rietdijk–Putnam–Penrose conventionality disputes within a neutral Block Universe framework.

7. **Coherence-Bounded Operational Locality: Simultaneous Phase Synchronization vs. Bell Correlations (Paper 10):** Vindicatory reconciliation of Bell's theorem (S_{CHSH}) with finite-latency relativistic causality ($\mathcal{S}_{\text{sim}}$), introduction of $\mathcal{D}_{\text{loc}}$, and the Delayed-Feedback Entangled Optical Clock protocol.

8. **Dissipative Quantum Reference Frames: Non-Unitary Transformations Beyond $\mathcal{L}_{\text{sim}}$ (Paper 11):** Generalization of unitary QRF shifts to open-system CPTP channels $\Lambda_{A\to B}$ weighted by $\mathcal{D}_{\text{loc}}$, QRF transformation fidelity bound $\mathcal{F}_{\text{QRF}}$, and distributed quantum internet architecture.

9. **Null-Hypothesis Calibration for Long-Baseline Quantum Gravity Interferometry (Paper 12):** Derivation of the Classical Synchronization Null-Hypothesis Visibility curve $V_{\text{null}}(\Omega, \Delta\ell)$ and Two-Tier Ξ_{QG} anomaly isolation protocol for MAGIS-100, AION, and MIGA.

10. **Thermodynamic Dissolution of Operational Simultaneity Across Gravitational Horizons (Paper 13):** Curved-spacetime redshift modulation $\mathcal{L}_{\text{sim}}(r) = \mathcal{L}_{\text{sim},\infty}\sqrt{1 - r_s/r}$, pre-horizon dissolution radius $r_{\text{diss}} > r_s$, resolution of Horizon Complementarity, and galactic Pulsar Timing Array ($\mathcal{D}_{\text{loc}} \to 0$) classification.

2 Architectural Synthesis: What the Metrics Mean and How We Use Them

To bridge microscopic open-quantum dynamics with macroscopic relativistic spacetimes without dimensional ambiguity, the framework operates through a four-tier coupled pipeline (implemented interactively in the `aeon-t.html` 14-metric diagnostic engine). *Notation Harmonization Note:* In v2.0, uppercase N_{eff} is strictly preserved for Surviving Coherent Cycles (Part I, Sec. 2), while lowercase n_{eff} denotes the Effective Group Refractive Index of optical transmission media (Papers 9–13).

2.1 Tier 1: Local Relativistic Clock Kinematics & Quantum Coherence

First, we anchor local clock rates to a covariant Zero-Angular-Momentum Observer (ZAMO) foliation using the relational dilation factor ρ_τ, clock rate delta $\Delta\rho_\tau = \rho_\tau - 1$, and Compton suppression H_Q. Over proper evolution time t_{pre}, environmental Caldeira–Leggett scattering (Γ_{th}) and gravitational self-energy dephasing (Γ_{grav}) accumulate into total decoherence rate $\Gamma_{\text{tot}} = \Gamma_{\text{th}} + \Gamma_{\text{grav}}$ and dimensionless action $\Omega = \Gamma_{\text{tot}} t_{\text{pre}}$.

The resulting **Timeline Integrity** $\Phi_\tau(\Omega) = \Phi_{\text{floor}} + (1 - \Phi_{\text{floor}})e^{-\Omega}$ measures the surviving local phase purity of a quantum clock, bounded below by the holographic Bekenstein–Hawking floor Φ_{floor} and above by unitarity. From Φ_τ, we compute usable coherent cycles $N_{\text{eff}} = (f_{\text{local}} t_{\text{pre}})\Phi_\tau$ and verify thermodynamic admissibility via the Lloyd–Bekenstein ratio $\eta_{\text{Lloyd}} \leq 1$.

2.2 Tier 2: Geometric Purity, Superradiance & Semiclassical Validity

Second, we evaluate the background spacetime geometry and semiclassical consistency. The **Weyl Conformal Purity** χ quantifies vacuum distortion induced by trace anomalies, while the Buckingham-Π ratio $R_{\text{DP}} = \Gamma_{\text{grav}}/\Gamma_{\text{th}}$ identifies whether a mesoscopic system sits in the thermal-decoherence ($R_{\text{DP}} < 1$) or gravitational-collapse ($R_{\text{DP}} > 1$) regime. For rotating black holes, $E_{\text{Penrose}}(\theta)$ and $(\alpha_G, \tau_{\text{inst}})$ track rotational energy extraction and superradiant cloud growth. Across all regimes, the L_∞ Chebyshev **Semiclassical Breakdown Index** $\Xi_{\text{QG}} \in (0, 1]$ acts as a real-time safety monitor against Planck-scale curvature, horizon blueshift, extremality, or thermal breakdown.

2.3 Tier 3: Operational Simultaneity Horizons & Coherence-Bounded Locality

Third, Papers 9 and 10 extend local clock coherence $(\rho_\tau, \Phi_\tau, \Gamma_{\text{tot}})$ across spatially separated networks of proper baseline $\Delta\ell$ and group index n_{eff}. Establishing active, real-time simultaneity requires a two-way Einstein–Poincaré electromagnetic handshake taking $\Delta\tau_{\text{sig}} = \frac{2\Delta\ell}{c n_{\text{eff}}}$. Comparing this signaling latency against the coherence window $\tau_{\text{coh}} = \rho_\tau/\Gamma_{\text{tot}}$ defines the **Operational Simultaneity Horizon**:

$$\mathcal{L}_{\text{sim}} = \frac{c\, n_{\text{eff}}\, \rho_\tau}{2\Gamma_{\text{tot}}}. \tag{1}$$

This horizon cleanly separates passive statistical correlations from active causal control:

- **Post-Selected Bell Correlations (S_{CHSH}):** Depend solely on local purity $\Phi_\tau(\Omega)$ via $S_{\text{CHSH}}(\Omega) = 2\sqrt{2}\,\Phi_\tau(\Omega)$, remaining invariant under spatial separation $\Delta\ell$ within a neutral four-dimensionalist manifold.
- **Operational Simultaneity Fidelity ($\mathcal{S}_{\text{sim}}$):** Measures real-time phase-locking fidelity without post-selection, decaying exponentially via $\mathcal{S}_{\text{sim}}(\Omega, \Delta\ell) = \Phi_\tau(\Omega)\exp(-\Delta\ell/\mathcal{L}_{\text{sim}})$.
- **Spatial Locality Decay Factor ($\mathcal{D}_{\text{loc}}$):** Isolates pure relativistic signaling-decoherence decay via $\mathcal{D}_{\text{loc}} = \frac{2\sqrt{2}\,\mathcal{S}_{\text{sim}}}{S_{\text{CHSH}}} = \exp(-\Delta\ell/\mathcal{L}_{\text{sim}})$, replacing idealized *naive classical point-locality* ($\mathcal{D}_{\text{loc}} \equiv 1$) with **Coherence-Bounded Operational Locality**.

2.4 Tier 4: Quantum Reference Frames, Null Calibration & Gravitational Horizons

Fourth, Papers 11–13 deploy $\mathcal{D}_{\text{loc}}$ and $\mathcal{L}_{\text{sim}}$ as operational diagnostic tools across three frontiers:

1. **Dissipative Quantum Reference Frames (Paper 11):** Spatial transformations between separated quantum clocks are governed by the non-unitary CPTP channel $\Lambda_{A\to B}(\rho_A) = \mathcal{D}_{\text{loc}}(\hat{U}_{A\to B}\rho_A\hat{U}^\dagger_{A\to B}) + (1-\mathcal{D}_{\text{loc}})\frac{\mathbb{I}}{d}$, bounding QRF fidelity by $\mathcal{F}_{\text{QRF}}$ and partitioning quantum networks into local active cores ($\Delta\ell \ll \mathcal{L}_{\text{sim}}$) and post-selected links ($\Delta\ell \gg \mathcal{L}_{\text{sim}}$).

2. **Quantum Gravity Interferometry Calibration (Paper 12):** $V_{\text{null}}(\Omega, \Delta\ell) = \Phi_\tau(\Omega)\mathcal{D}_{\text{loc}}$ provides the exact classical relativistic null-hypothesis visibility curve for long-baseline interferometers (MAGIS-100, AION, MIGA), enabling the Two-Tier Ξ_{QG} monitor to reject false-positive gravitational decoherence claims.

3. **Gravitational Horizons & Galactic Timing Arrays (Paper 13):** In curved spacetime, gravitational redshift contracts the local horizon via $\mathcal{L}_{\text{sim}}(r) = \mathcal{L}_{\text{sim},\infty}\sqrt{1-r_s/r}$, dissolving operational simultaneity smoothly at $r_{\text{diss}} > r_s$ before the event horizon is reached and placing galactic Pulsar Timing Arrays in the asymptotic $\mathcal{D}_{\text{loc}} \to 0$ post-selected regime.

3 Unified Dimensionless Symbol & Metric Glossary (v2.0)

Symbol	Name	Defining Equation	Domain & Physical Interpretation
ρ_τ	ZAMO Dilation Factor	$\frac{\gamma_{\text{pre}}/\sqrt{N^2_{\text{pre}}(\theta)}}{\gamma_{\text{obs}}/\sqrt{1-C_{\text{obs}}}}$	$(0,\infty)$; ratio of observer proper time $d\tau_{\text{obs}}$ to target proper time $d\tau_{\text{pre}}$.
$\Delta\rho_\tau$	Clock Rate Delta	$\rho_\tau - 1$	$(-1,\infty)$; net fractional relativistic clock drift ($z = \Delta\rho_\tau$).
H_Q	ZAMO Back-Reaction Factor	$\sqrt{N^2_{\text{pre}}(\theta)}/\gamma_{\text{pre}}$	$(0,1]$; combined gravitational and kinematic suppression of Compton frequency.
Ω	Decoherence Action	$(\Gamma_{\text{th}} + \Gamma_{\text{grav}}) \cdot t_{\text{pre}}$	$[0,\infty)$; cumulative environmental and gravitational phase-scattering depth ($\Gamma_{\text{tot}} = \Gamma_{\text{th}} + \Gamma_{\text{grav}}$).
Φ_{floor}	Holographic Coherence Floor	$(1 + \pi \cdot 10^{2\zeta})^{-1}$	$(0,(1+\pi)^{-1}]$; inverse Bekenstein–Hawking Hilbert-space dimension ($\zeta = \log_{10}(L/\ell_P)$).
$\Phi_\tau(\Omega)$	Timeline Integrity	$\Phi_{\text{floor}} + (1-\Phi_{\text{floor}})e^{-\Omega}$	$[\Phi_{\text{floor}}, 1]$; surviving local quantum state purity bounded by holographic unitarity.
N_{eff}	Surviving Coherent Cycles	$(f_{\text{local}} \cdot t_{\text{pre}}) \cdot \Phi_\tau$	$[0,\infty)$; total usable phase-coherent quantum oscillations prior to thermalization.
η_{Lloyd}	Lloyd–Bekenstein Ratio	$N_{\text{eff}}/(S_{\text{BH}}/k_B)$	$[0,\infty)$; coherent cycle utilization relative to holographic bit capacity (≤ 1 bounded).
χ	Weyl Conformal Purity	$(1+60A_W)^{-1} = (1 + C^2_{\text{clamped}}/\pi^2)^{-1}$	$[\pi^2/(\pi^2+4), 1]$; normalized static vacuum purity under conformal trace anomaly.
R_{DP}	Gravity-to-Thermal Ratio	$\Gamma_{\text{grav}}/\Gamma_{\text{th}} = (M_{\text{eff}}/M_P)^2({}_{\text{th}}/L)^3$	$[0,\infty)$; ratio of Penrose–Diósi collapse rate to Caldeira–Leggett scattering rate.
$E_{\text{Penrose}}(\theta)$	Penrose–Zel'dovich Gain	$\max(0, 1 - f_{\text{local}}/f_H)\sin^2\theta \cdot \eta_{\text{NT}}(r_{\text{ISCO}})$	$[0, 0.4226]$; latitude-dependent rotational energy extraction efficiency.

Symbol	Name	Defining Equation	Domain & Physical Interpretation
$\alpha_G, \tau_{\text{inst}}$	Grav. Atom Coupling & Growth	$\alpha_G = \frac{M_{\text{eff}} m_0}{M_P^2}$, $\tau_{\text{inst}} = \frac{24 r_g (1+\chi_s)}{c a_* \alpha_G^9}$	Dimensionless coupling and Detweiler $2p$ ($\ell = m = 1$) cloud instability timescale.
Ξ_{QG}	Semiclassical Breakdown Index	$\max\left(\frac{\ell_P}{L}, \frac{10^{-6}}{N^2(\theta)}, \frac{a_*^2+Q_*^2}{0.9998}, \frac{T_{\text{eff}}}{T_P}\right)$	$(0,1]$; two-tier L_∞ Chebyshev regulator monitoring proximity to quantum-gravity breakdown walls.
n_{eff}	Effective Group Refractive Index	c/v_g	$[1,\infty)$; waveguide group refractive index ($n_{\text{eff}} = 1$ in vacuum; ≈ 1.468 in telecom fiber).
$\Delta\tau_{\text{sig}}$	Round-Trip Signal Delay	$\frac{2\Delta\ell}{c\, n_{\text{eff}}}$	$[0,\infty)$; two-way Einstein–Poincaré synchronization latency across proper baseline $\Delta\ell$.
$\mathcal{L}_{\text{sim}}$	Operational Simultaneity Horizon	$\frac{c\, n_{\text{eff}}\, \rho_\tau}{2\Gamma_{\text{tot}}}$	$(0,\infty)$; maximum baseline for active real-time closed-loop phase synchronization.
$\mathcal{S}_{\text{sim}}(\Omega, \Delta\ell)$	Operational Simultaneity Fidelity	$\Phi_\tau(\Omega) \cdot \exp(-\Delta\ell/\mathcal{L}_{\text{sim}})$	$[0,1]$; actionable real-time phase-locking fidelity without post-selection.
$S_{\text{CHSH}}(\Omega)$	Post-Selected Bell Correlation	$2\sqrt{2} \cdot \Phi_\tau(\Omega)$	$[0, 2\sqrt{2}]$; post-selected CHSH correlation independent of $\Delta\ell$ (> 2 violates classical bound).
$\mathcal{D}_{\text{loc}}$	Spatial Locality Decay Factor	$\frac{2\sqrt{2}\,\mathcal{S}_{\text{sim}}}{S_{\text{CHSH}}} = \exp\left(-\frac{\Delta\ell}{\mathcal{L}_{\text{sim}}}\right)$	$(0,1]$; dimensionless decay of operational point-locality across baseline $\Delta\ell$.
$\Lambda_{A\to B}$	Coherence-Bounded QRF Channel	$\mathcal{D}_{\text{loc}}(\hat{U}_{A\to B}\rho_A\hat{U}_{A\to B}^\dagger) + (1 - \mathcal{D}_{\text{loc}})\frac{\mathbb{I}}{d}$	CPTP map governing non-unitary Quantum Reference Frame shifts across $\Delta\ell$.
$\mathcal{F}_{\text{QRF}}$	QRF Transformation Fidelity	$\frac{1}{d} + \left(1 - \frac{1}{d}\right)\mathcal{D}_{\text{loc}}$	$[1/d, 1]$; upper bound on state fidelity under spatially separated QRF transformation.
$V_{\text{null}}(\Omega, \Delta\ell)$	Null-Hypothesis Visibility	$\Phi_\tau(\Omega) \cdot \mathcal{D}_{\text{loc}}$	$[0,1]$; classical synchronization-lag visibility benchmark for long-baseline interferometry.
$\mathcal{L}_{\text{sim}}(r)$	Redshift-Modulated Horizon	$\mathcal{L}_{\text{sim},\infty}\sqrt{1 - r_s/r}$	$[0, \mathcal{L}_{\text{sim},\infty})$; local proper simultaneity horizon in Schwarzschild geometry ($r > r_s$).
r_{diss}	Pre-Horizon Dissolution Radius	$r_s\left[1 - \left(\frac{\Delta\ell_{\text{link}}}{\mathcal{L}_{\text{sim},\infty}}\right)^2\right]^{-1}$	(r_s, ∞); critical radius where $\mathcal{L}_{\text{sim}}(r_{\text{diss}}) = \Delta\ell_{\text{link}}$ and active simultaneity dissolves.

4 Verified Foundational Bibliography & Acknowledgments

Authorship & AI Attribution: Authored by **Luís Pimenta Damásio** (Independent Researcher, Lisbon, Portugal; ORCID: 0009-0001-7606-7557) in transparent scientific collaboration with Google's **Gemini** AI assistant, which assisted with symbolic derivation verification, LaTeX/PDF typesetting, and bibliographic cross-referencing.

1. Almheiri, A., Engelhardt, N., Marolf, D., & Maxfield, H. (2019). The entropy of bulk quantum fields and the entanglement wedge of an evaporating black hole. *J. High Energy Phys.*, 2019(12), 063.

2. Arnowitt, R., Deser, S., & Misner, C. W. (1962). The dynamics of general relativity. In *Gravitation: An Introduction to Current Research* (Ch. 7, pp. 227–265). John Wiley & Sons.

3. Bardeen, J. M., Press, W. H., & Teukolsky, S. A. (1972). Rotating black holes: Locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *Astrophys. J.*, 178, 347–370.

4. Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1(3), 195–200.

5. Caldeira, A. O., & Leggett, A. J. (1983). Path integral approach to quantum Brownian motion. *Physica A*, 121(3), 587–616.

6. Clauser, J. F., Horne, M. A., Shimony, A., & Holt, R. A. (1969). Proposed experiment to test local hidden-variable theories. *Phys. Rev. Lett.*, 23(15), 880–884.

7. Detweiler, S. (1980). Klein–Gordon equation and rotating black holes. *Phys. Rev. D*, 22(10), 2323–2326.

8. DeWitt, B. S. (1965). *Dynamical Theory of Groups and Fields*. Gordon and Breach.

9. Diósi, L. (1989). Models for universal reduction of macroscopic quantum fluctuations. *Phys. Rev. A*, 40(3), 1165–1174.

10. Einstein, A. (1905). Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10), 891–921.

11. Giacomini, F., Castro-Ruiz, E., & Brukner, Č. (2019). Quantum mechanics and the covariance of physical laws in quantum reference frames. *Nat. Commun.*, 10(1), 494.

12. Gorini, V., Kossakowski, A., & Sudarshan, E. C. G. (1976). Completely positive dynamical semigroups of N-level systems. *J. Math. Phys.*, 17(5), 821–825.

13. Hawking, S. W. (1975). Particle creation by black holes. *Commun. Math. Phys.*, 43(3), 199–220.

14. Lindblad, G. (1976). On the generators of quantum dynamical semigroups. *Commun. Math. Phys.*, 48(2), 119–130.

15. Lloyd, S. (2000). Ultimate physical limits to computation. *Nature*, 406(6799), 1047–1054.

16. Margolus, N., & Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D*, 120(1–2), 188–195.

17. Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman.

18. Page, D. N. (1993). Information in black hole radiation. *Phys. Rev. Lett.*, 71(23), 3743–3746.

19. Page, D. N., & Wootters, W. K. (1983). Evolution without evolution: Dynamics described by stationary observables. *Phys. Rev. D*, 27(12), 2885–2892.

20. Penington, G. (2020). Entanglement wedge reconstruction and the information paradox. *J. High Energy Phys.*, 2020(09), 002.

21. Penrose, R. (1989). *The Emperor's New Mind*. Oxford University Press.

22. Penrose, R. (1996). On gravity's role in quantum state reduction. *Gen. Relativ. Gravit.*, 28(5), 581–600.

23. Putnam, H. (1967). Time and physical geometry. *J. Philos.*, 64(8), 240–247.

24. Rietdijk, C. W. (1966). A rigorous proof of determinism derived from the special theory of relativity. *Philos. Sci.*, 33(4), 341–344.

25. Teukolsky, S. A. (1973). Perturbations of a rotating black hole. I. *Astrophys. J.*, 185, 635–647.

26. Thorne, K. S., Price, R. H., & MacDonald, D. A. (1986). *Black Holes: The Membrane Paradigm*. Yale University Press.